

Density Matrix Classifier

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Abstract

We introduce a new type of quantum machine learning classifier based on the estimation of Frobenius distance between density matrices. The central idea is to encode all samples in the dataset belonging to a particular class into a corresponding density matrix using the quantum feature map methodology. The same quantum feature map is used to encode every new test sample into its own corresponding density matrix. Frobenius distances between the sample density matrix and the density matrices encoding all classes present in the training dataset are calculated. The class label is assigned by selecting the class with minimum distance. The density matrix classifier uses parameterised quantum circuits. On the Glass Identification dataset, it performed on par with, or sometimes exceeded, performance of standard classical classifiers including multi-layer perceptron, random forest, support vector machine, and logistic regression. The model is resistant to certain types of noise and is suitable for execution on NISQ computers.

1 Introduction

The field of quantum machine learning (QML) has experienced significant growth in recent years [1] primarily driven by two factors. Firstly, many QML algorithms can be executed successfully on current noisy intermediate-scale quantum (NISQ) [2] computers due to their resistance to certain types of noise. This noise resistance may result from either the application of variational methods used for training QML models [3, 4, 5], or through working with expectation values where averaging across large number of measurements helps to suppress the variance [6]. The bias can be (partially) suppressed through various error mitigation techniques [7]. Secondly, parameterised quantum circuits (PQC) have exponentially larger expressive power than comparable classical neural networks for certain applications [8, 9]. The expressive power of PQCs has been extensively studied on a number of use cases that involved experiments on both quantum simulators and the cutting-edge quantum computers available at the time [10, 11].

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The suite of QML tools now includes discriminative and generative models that are direct counterparts to the corresponding classical machine learning models. Discriminative models include quantum neural networks [12] and quantum kernels [13] that can be compared with classical feed-forward neural networks and support vector machines [14] respectively. Generative models like quantum generative adversarial networks (qGAN) [15] and quantum circuit Born machines (QCBM) [16] can be compared, respectively, with classical GANs [17] and restricted Boltzmann machines (RBM) [18].

In this paper we extend the range of available discriminative QML models by adding a new classifier based on two key concepts: quantum feature maps and Frobenius distance between density matrices. A quantum feature map consists of encoding each sample from a given dataset into its corresponding quantum state with the help of a PQC. A low-dimensional sample can be mapped into a quantum state – a state vector in a high-dimensional Hilbert space. It can also be described by the corresponding density matrix. All samples belonging to a particular class can be represented by the class density matrix which is an equally weighted linear combination of the individual sample density matrices (equally weighted statistical ensemble of all samples belonging to the class). The classification of a new sample is then performed by measuring the distances between the density matrix representing the sample and each of the density matrices representing all of the classes in the dataset. The sample is assigned to the class corresponding to the shortest distance. There are many possible ways in which these distances can be measured, the most suitable being the Frobenius distance [19].

This density matrix classifier provides the following benefits:

- No iterative training methods are required. Density matrices are created for each class. The number of operations scales linearly with the number of training data points.
- Updating the density matrices to include new information (i.e., new training data samples) is a simple process, scaling linearly with the number of new training samples.
- Only expectation values need to be calculated using the quantum computer. This is more robust to noise and to scaling in the number of qubits utilised, as compared to algorithms which measure and use bitstrings directly. When calculating expectations, bit flip errors get averaged away. In contrast, algorithms which calculate utility value by measuring bitstrings and inserting them into an objective function can obtain very different values if any bits within the bitstring contain errors.

The paper is organised as follows. In Section 2 we introduce the concepts of quantum feature maps and Frobenius distance between density matrices. In Section 3 we describe the proposed density matrix classifier and present results of numerical experiments. In Section 4 we conclude and outline directions for further research.

2 Quantum Feature Map and Frobenius Distance

Let $\mathcal{D}$ be a dataset consisting of samples belonging to N non-overlapping subsets (classes) $\mathcal{C}_1, \dots, \mathcal{C}_N$:

$$\mathcal{C}_1 \cup \dots \cup \mathcal{C}_N = \mathcal{D}, \quad \mathcal{C}_i \cap \mathcal{C}_j = \emptyset \quad \forall i \neq j. \quad (1)$$

Each sample in the dataset, $\mathbf{x} \in \mathcal{D}$, can be represented as a vector of features, $\mathbf{x} := (x_1, \dots, x_m)^\top$, where m is the number of features, or the *dimensionality* of the dataset. We define the feature map φ as a function that maps samples $\mathbf{x}$ into operators $\varphi(\mathbf{x})$ of a Hilbert space $\mathcal{H}$. The feature map φ is realised on a quantum computer via a unitary transformation $U(\mathbf{x})$ applied to the initial state $|\mathbf{0}\rangle \equiv |0\rangle^{\otimes n}$:

$$\varphi(\mathbf{x}) := U(\mathbf{x}) |\mathbf{0}\rangle \langle \mathbf{0}| U(\mathbf{x})^\dagger \equiv |\mathbf{x}\rangle \langle \mathbf{x}|. \quad (2)$$

Note that if the transformation φ is specified on n qubits, then the dimension of the Hilbert space $\mathcal{H}$ of linear operators is $4^n = 2^n \times 2^n$.

The quantum state $|\mathbf{x}\rangle := U(\mathbf{x}) |\mathbf{0}\rangle$ is a *pure* quantum state encoding sample $\mathbf{x}$, and the feature map $\varphi(\mathbf{x})$ is the corresponding density matrix. The density matrix formalism allows us to describe not only pure but also *mixed* quantum states, i.e., statistical ensembles of pure quantum states. Thus, the density matrix approach can be used to encode not just a single sample but the whole subset of samples belonging to a particular class.

For each class $\mathcal{C}_i \subset \mathcal{D}$, $i = 1, \dots, N$, we construct a density matrix ρ_i defined as an equally weighted statistical ensemble of all samples in $\mathcal{C}_i$:

$$\rho_i = \frac{1}{[\mathcal{C}_i]} \sum_{k=1}^{[\mathcal{C}_i]} |\mathbf{x}^k\rangle \langle \mathbf{x}^k|, \quad (3)$$

where $[\mathcal{C}_i]$ denotes the cardinality of $\mathcal{C}_i$ and the sum goes over all samples $\mathbf{x}^k$, $k = 1, \dots, [\mathcal{C}_i]$ in class $\mathcal{C}_i$.

We can estimate the *closeness* of any two classes, $\mathcal{C}_i$ and $\mathcal{C}_j$, as the distance between the corresponding density matrices ρ_i and ρ_j . There are many possible ways to define the distance between density matrices [20, 21]. For example:

- Jensen-Shannon distance:

$$d_J(\rho_i, \rho_j) = \sqrt{S\left(\frac{\rho_i + \rho_j}{2}\right) - \frac{1}{2}S(\rho_i) - \frac{1}{2}S(\rho_j)},$$

where

$$S(\rho) = -\text{Tr}(\rho \ln \rho)$$

is the von Neumann entropy.

- Bures distance:

$$d_B(\rho_i, \rho_j) = \sqrt{1 - \sqrt{F(\rho_i, \rho_j)}} ,$$

where

$$F(\rho_i, \rho_j) = (\text{Tr} \sqrt{\sqrt{\rho_i} \rho_j \sqrt{\rho_i}})^2$$

is the fidelity of two quantum states.

- Hellinger distance:

$$d_H(\rho_i, \rho_j) = \sqrt{1 - \text{Tr}(\sqrt{\rho_i} \sqrt{\rho_j})} .$$

However, the distance that we will utilise in the experiments that follow is the Frobenius distance:

$$d_F(\rho_i, \rho_j) = \frac{1}{2} \sqrt{\text{Tr}[(\rho_i - \rho_j)^2]} . \quad (4)$$

The primary motivation for choosing the Frobenius distance over other alternatives is that it can be efficiently estimated on a quantum computer [19], as explained in Appendix C.

3 Density Matrix Classifier

3.1 DMC Definition

We now have all the necessary ingredients to construct a classifier. The density matrix classifier (DMC) is a lazy classifier in the spirit of the K -nearest neighbours model [14] since it does not require training and all information about the dataset is encoded in the density matrices. It also has some similarities with support vector machines due to the feature map which takes low-dimensional samples into objects in a high-dimensional Hilbert space.

The DMC can be formulated as the following algorithm:

- 1) Start with the dataset consisting of N non-overlapping subsets (classes) $\mathcal{C}_1, \dots, \mathcal{C}_N$ as described in (1). Each sample $\mathbf{x}$ in the dataset is an m -dimensional vector of real-valued features. Discrete or categorical variables can also be represented.
- 2) Specify an n -qubit quantum circuit that implements the unitary transformation $U(\mathbf{x})$ applied to the initial all-zero state $|0\rangle^{\otimes n}$. For example, the quantum circuit may consist of several layers of adjustable 1-qubit gates encoding components of $\mathbf{x}$ into qubit rotation angles, and layers of fixed 2-qubit gates creating entanglement.
- 3) For each class $\mathcal{C}_i, i = 1, \dots, N$, and for each sample in class $\mathcal{C}_i, |\mathbf{x}^k\rangle, k = 1, \dots, [\mathcal{C}_i]$, construct the feature $\varphi(\mathbf{x}^k)$ given by expression (2) using the unitary transformation $U(\mathbf{x}^k)$.

- 4) For each class $\mathcal{C}_i, i = 1, \dots, N$, construct a density matrix ρ_i defined as an equally weighted statistical ensemble of all samples in $\mathcal{C}_i$ as per expression (3).
- 5) For each new sample to be classified, $\mathbf{x}$, construct its own feature map $\varphi(\mathbf{x})$ and density matrix ρ using the unitary transformation $U(\mathbf{x})$. For each class $\mathcal{C}_i, i = 1, \dots, N$, calculate the Frobenius distance $d_F(\rho_i, \rho)$ given by expression (4). The new sample is assigned a class label corresponding to the class with the smallest Frobenius distance.

Examples of feature encoding and quantum circuits used to implement feature maps are discussed in Section 3.2.2, and illustrated in Figures 2–3 and Figures 4–5 respectively.

3.2 Numerical Experiments

We conducted experiments on Ankaa-2 – the 4th generation superconducting quantum processor released by Rigetti Computing in 2023 [22]. Performance of the DMC was benchmarked against several standard classical models using the Glass Identification dataset [23].

3.2.1 Dataset

The dataset consists of 214 samples, with each glass sample described by nine continuous features (Refractive Index, Sodium, Magnesium, Aluminium, Silicon, Potassium, Calcium, Barium, Iron) and four class labels (Window, Container, Tableware, Headlamp). The dataset is heavily imbalanced, which makes it harder to find the right balance between precision and recall and prompts us to choose the F1 score with ‘macro’ averaging across classes as the preferred metric. Table 1 shows the distribution of samples by classes.

Class label	Number of samples
Window	163
Container	13
Tableware	9
Headlamp	29

Table 1: Distribution of samples by classes for the Glass Identification dataset.

Feature importance analysis conducted with the help of a random forest classifier suggests the ranking of features in terms of their relative predictive power as shown in Table 2.

Feature	Importance
Magnesium (Mg)	21.5%
Barium (Ba)	18.9%
Aluminium (Al)	13.6%
Sodium (Na)	12.5%
Potassium (K)	10.7%
Calcium (Ca)	8.1%
Refractive Index (RI)	6.9%
Silicon (Si)	6.4%
Iron (Fe)	1.4%

Table 2: Feature importance for the Glass Identification dataset, as determined by a random forest classifier.

We performed classification using all nine features, as well as the top three features obtained from the feature importance analysis. The reasons for experimenting with only the three most important features are given in the following subsection.

3.2.2 Feature Encoding and Quantum Feature Map

Feature encoding is performed using the *angle encoding* scheme. Each feature $x_i, i = 1, \dots, m$ of a sample $\mathbf{x} := (x_1, \dots, x_m)^\top$ is mapped into the corresponding *rotation angle* – the adjustable parameter of a 1-qubit gate. Each 1-qubit gate operator (mathematically, a 2×2 unitary matrix) can be seen as a rotation of the qubit state (a point on the unit sphere called the Bloch sphere) around some axis. In practice, and without loss of generality, we work with rotations around the x , y , and z axis (with z -basis being the computational basis). Any point on the Bloch sphere can be reached from any other point on the Bloch sphere by just two rotations around any two orthogonal axes.

Given the dataset $\mathcal{C}$ with the total number of samples equal to the dataset cardinality $|\mathcal{C}|$, for each sample $\mathbf{x}^k, k = 1, \dots, |\mathcal{C}|$, the mapping of the feature x_i^k to the corresponding rotation angle θ_i^k is performed as follows:

$$\theta_i^k = \alpha\pi \frac{x_i^k - \min_{j=1, \dots, |\mathcal{C}|} x_i^j}{\max_{j=1, \dots, |\mathcal{C}|} x_i^j - \min_{j=1, \dots, |\mathcal{C}|} x_i^j}, \quad (5)$$

where α is the range parameter. In our experiments we set $\alpha = 1/4$ and, therefore, we have $\theta_i^k \in [0, \pi/4], \forall i, k$.

However, before the feature is mapped into a rotation angle of a 1-qubit gate, it should be pre-processed to even out its distribution. If the feature is distributed more or less evenly, the mapping scheme (5) works well without any pre-processing. But if the feature values are concentrated around the minimum or the maximum values, we need to stretch the distribution up from the minimum or down from the

maximum in order to enhance the discriminatory power of a quantum feature map. Intuitively, we can understand it as an attempt to reach as many regions of the Hilbert space as possible and increase the distance between different data points. In our experiments, we adopted the following simple feature pre-processing rule:

- **If** $\text{median} - \min < \frac{1}{4}(\max - \min)$ **Then** $x \rightarrow x^\gamma$, $\gamma = \frac{1}{2}$.
- **If** $\frac{1}{4}(\max - \min) \leq \text{median} - \min \leq \frac{3}{4}(\max - \min)$ **Then** $x \rightarrow x^\gamma$, $\gamma = 1$.
- **If** $\text{median} - \min > \frac{3}{4}(\max - \min)$ **Then** $x \rightarrow x^\gamma$, $\gamma = \frac{3}{2}$.

The QPU graph is a 3×3 square grid of computational qubits connected by tunable couplers. This graph can be represented by a section of the 84-qubit Ankaa-2 system, or a full 9-qubit Novera system [24] – a 9-qubit version of the Ankaa architecture.

Figures 2 and 3 in Appendix A show the feature embeddings on the QPU graph for, respectively, the nine features scenario and the top three features scenario. Figures 4 and 5 in Appendix B display the corresponding quantum feature maps. In both scenarios we have repeat encoding of the features using the same qubits, between layers of 2-qubit gates creating entanglement.

The top three features scenario differs from the nine features scenario in two important respects. Firstly, the top three features dataset is harder to classify since the classifier is restricted to utilise limited information. Secondly, the same 9-qubit QPU graph is used for multiple qubit encoding: each feature is encoded in three qubits. This should provide the DMC with an opportunity to demonstrate the expressive power of quantum feature maps in outperforming the classical benchmarks.

The hardness of a classification problem increases with the reduction of a number of features with high predictive power, as less information is available to perform the classification. Although this affects both classical and quantum models, the larger expressive power of quantum models may help to preserve more information and use it in a more efficient way, especially when this information is shared across multiple qubits. The ability of a quantum model to learn complex functions better than classical models from small datasets, and perform similarly to classical models when larger training datasets become available is well known, and has been experimentally established on, for example, a discrete log dataset [25].

3.2.3 Classical Benchmarks

The classical benchmarks of choice are four standard classical classifiers which are widely used in data science applications: multi-layer perceptron (MLP), random forest (RF), support vector machine (SVM), and logistic regression (LR). Collectively, they cover all the main classical ML approaches to classification problems.

MLP is a *feedforward neural network* trained through the backpropagation of error with gradient descent. RF is an *ensemble learning model* that builds a strong classifier through the aggregation of predictions made by several individual weak classifiers (decision trees) trained in parallel on a random subset of the data. SVM

perform classification using the *kernel trick* to implicitly map their inputs into high-dimensional feature spaces where linear classification can be performed more easily. LR is, arguably, the most basic and simple model consisting of mapping a linear combination of features into the interval $[0, 1]$ (which can be interpreted as probability) with the help of a *logistic function*.

We use the scikit-learn implementation of these classifiers [26] with the set of optimised parameters as shown in Table 3.

scikit-learn classifier	parameter value
Logistic Regression Classifier	solver = 'lbfgs' penalty = 'l2' C = 10 max_iter = 5000 random_state = 0
MLP Classifier	solver = 'adam' hidden_layer_sizes = (50, 50) activation = 'relu' alpha = 0.1 max_iter = 5000 random_state = 0
Random Forest Classifier	criterion = 'entropy' n_estimators = 500 random_state = 0
Support Vector Classifier	kernel = 'rbf' C = 100 random_state = 0

Table 3: scikit-learn classifiers – optimised model parameters. All other model parameters were fixed at their default values [26].

Similar to the feature standardisation for DMC, where all features are mapped to the same rotation angle interval $[0, \alpha\pi]$ as per expression (5), classical classifiers such as MLP, SVM and LR also work with standardised features. For our experiments we use the scikit-learn StandardScaler class [27], which standardises features by removing their mean values and scaling them to unit variance.

3.2.4 Experimental Results

Figure 1 displays F1 score results for the classical benchmarks and the DMC executed on a quantum simulator, DMC (sim), and on Rigetti’s Ankaa-2 QPU, DMC (QPU). All F1 scores are out-of-sample: we used the scikit-learn test-train-split class [28] to generate ten random 80:20 splits into training and test datasets. The QPU results were obtained using the approximation based on the first order Pauli

expectation values (Appendix C). The quantum simulator results were obtained using both the first order Paulis approximation and the exact formula (4). The optimal value of the range parameter α was found to be 0.2 through a simple grid search method.

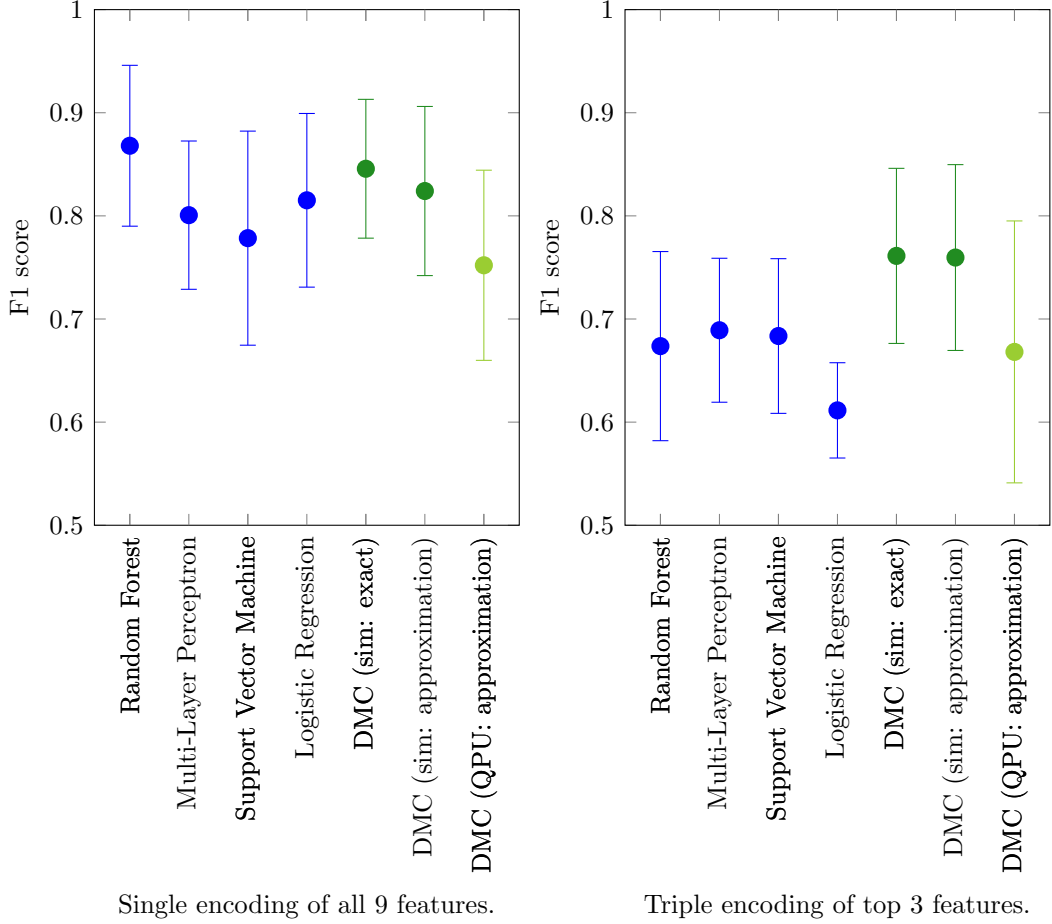


Figure 1: Out-of-sample F1 scores for classical (shown in blue) and quantum (shown in green) classifiers using the Glass Identification dataset. Error bars indicate ± 1 standard deviation. All F1 scores were calculated with 'macro' averaging across classes.

In Figure 1 the dots indicate the mean values and the error bars indicate the uncertainty in F1 scores (± 1 standard deviation). The nine features scenario with single qubit encoding has a higher average F1 score across all classifiers in comparison with the top three features scenario with triple qubit encoding. This is not surprising since nine features contain significantly more information as per the feature importance analysis presented in Table 2.

DMC (quantum simulator) performs at par with the classical benchmarks on the nine features scenario. At the same time it ranks above all the classical benchmarks on the top three features scenario with triple qubit feature encoding, where the mean DMC F1 score is almost a full one standard deviation above the mean F1 scores of all the classical classifiers. We can interpret this result as providing evidence of the

expressive power of the quantum feature map and ability of the DMC to preserve more valuable information. While it may not be clearly visible when we work with all nine features, dealing with the reduced set of features puts more pressure on the classifiers (in terms of limited available predictive information) and suppresses performance of the classical models more than the quantum DMC.

The QPU results are noisier than the results obtained on a quantum simulator and show lower average F1 scores. However, even with the noise present in the system, the QPU performs at par with the classical classifiers on the three features dataset – the larger expressive power of the quantum feature map may provide the necessary compensation.

4 Conclusions and Outlook

The proposed Density Matrix Classifier is a promising QML model that has the potential to outperform the standard classical classifiers on hard to classify datasets, where the larger expressive power of parameterised quantum circuits may play a crucial role. The model is suitable for execution on NISQ devices and can be improved and extended in several directions.

Firstly, improved fidelity of the individual qubits in combination with various noise reduction techniques should lead to the convergence of QPU and quantum simulator results.

Secondly, further experiments on finding an optimal circuit architecture may lead to improvements in the generalisation and approximation capabilities of the model. In this regard, the new results presented in [25] on exploring the non-Cliffordness of the quantum circuits may play a crucial role.

Finally, we note that the approximation based on the first order Pauli expectation values closely matches the exact value of the Frobenius distance for both feature scenarios considered in the paper. This implies potential scalability of the proposed algorithm – as discussed in Appendix C, using the exact formula based on direct calculation of the density matrix requires $\mathcal{O}(4^n)$ calculations for n qubits, while first order Paulis require $\mathcal{O}(3n)$ computations.

Bibliography

- [1] Antoine Jacquier and Oleksiy Kondratyev. *Quantum Machine Learning and Optimisation in Finance: On the Road to Quantum Advantage*. Packt, 2022.
- [2] John Preskill. Quantum Computing in the NISQ era and beyond. *Quantum*, 2, 2018.
- [3] Alberto Peruzzo, Jarrod McClean, Peter Shadbolt, Man-Hong Yung, Xiao-Qi Zhou, Peter J. Love, Alán Aspuru-Guzik, and Jeremy L. O’Brien. A variational eigenvalue solver on a photonic quantum processor. *Nature Communications*, 5(4213), 2014.

- [4] Abhinav Kandala, Antonio Mezzacapo, Kristan Temme, Maika Takita, Markus Brink, Jerry M. Chow, and Jay M. Gambetta. Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets. *Nature*, 549, 2017.
- [5] Nikolaj Moll, Panagiotis Barkoutsos, Lev S. Bishop, Jerry M. Chow, Andrew Cross, Daniel J. Egger, Stefan Filipp, Andreas Fuhrer, Jay M. Gambetta, Marc Ganzhorn, Abhinav Kandala, Antonio Mezzacapo, Peter Müller, Walter Riess, Gian Salis, John Smolin, Ivano Tavernelli, and Kristan Temme. Quantum optimization using variational algorithms on near-term quantum devices. *Quantum Science and Technology*, 3(3), 2018.
- [6] David Garvin, Oleksiy Kondratyev, Alexander Lipton, and Marco Painsi. *ADIA Lab Transactions (forthcoming)*, chapter Symmetric Encryption on a Quantum Computer. World Scientific, 2024.
- [7] Andrew Arrasmith, Andrew Patterson, Alice Boughton, and Marco Painsi. Development and demonstration of an efficient readout error mitigation technique for use in nisy algorithms. *arXiv:2303.17741*, 2023.
- [8] Marcello Benedetti, Erika Lloyd, Stefan Sack, and Mattia Fiorentini. Parameterized quantum circuits as machine learning models. *Quantum Science and Technology*, 4(4), 2019.
- [9] Yuxuan Du, Min-Hsiu Hsieh, Tongliang Liu, and Dacheng Tao. Expressive power of parameterized quantum circuits. *Physical Review Research*, 2(033125), 2020.
- [10] Oleksiy Kondratyev. Non-differentiable learning of quantum circuit born machine with genetic algorithm. *Wilmott*, 114, 2021.
- [11] Brian Coyle, Maxwell Henderson, Justin Chan Jin Le, Niraj Kumar, Marco Painsi, and Elham Kashefi. Quantum versus classical generative modelling in finance. *Quantum Science and Technology*, 6(2), 2021.
- [12] Edward Farhi and Hartmut Neven. Classification with quantum neural networks on near-term processors. *arXiv:1802.06002*, 2018.
- [13] Xinbiao Wang, Yuxuan Du, Yong Luo, and Dacheng Tao. Towards understanding the power of quantum kernels in the NISQ era. *Quantum*, 5(531), 2021.
- [14] Sebastian Raschka and Vahid Mirjalili. *Python Machine Learning*. Packt, 3rd edition, 2019.
- [15] Amin Assouel, Antoine Jacquier, and Alexei Kondratyev. A quantum generative adversarial network for distributions. *Quantum Machine Intelligence*, 4(2), 2022.
- [16] Jin-Guo Liu and Lei Wang. Differentiable learning of quantum circuit born machine. *Physical Review A*, 98(062324), 2018.

- [17] Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. *arXiv:1406.2661*, 2014.
- [18] Geoffrey Hinton. A practical guide to training restricted boltzmann machines. *Department of Computer Science, University of Toronto, UTML TR 2010-003*, 2010.
- [19] David Garvin, Oleksiy Kondratyev, Alexander Lipton, and Marco Painsi. Quantum two-sample test for investment strategies. *SSRN 4789400*, 2024.
- [20] Steffen Wissman, Bruno Leggio, and Heinz-Peter Breuer. Detecting initial system-environment correlations: Performance of various distance measures for quantum states. *Physical Review A*, 88(2), 2013.
- [21] Suvrit Sra. Metrics induced by Jensen-Shannon and related divergences on positive definite matrices. *Linear Algebra and its Applications*, 616, 2021.
- [22] Rigetti Computing. Rigetti Announces Public Availability of Ankaa-2 System with a 2.5x Performance Improvement Compared to Previous QPUs. [/https://www.globenewswire.com/news-release/2024/01/04/2804006/0/en/Rigetti-Announces-Public-Availability-of-Ankaa-2-System-with-a-2-5x-Performance-Improvement-Compared-to-Previous-QPUs.html](https://www.globenewswire.com/news-release/2024/01/04/2804006/0/en/Rigetti-Announces-Public-Availability-of-Ankaa-2-System-with-a-2-5x-Performance-Improvement-Compared-to-Previous-QPUs.html), 2024.
- [23] B. German. Glass Identification. UCI Machine Learning Repository, 1987.
- [24] Rigetti Computing. Rigetti Launches the Novera QPU, the Company’s First Commercially Available QPU. [/https://www.globenewswire.com/news-release/2023/12/06/2792153/0/en/Rigetti-Launches-the-Novera-QPU-the-Company-s-First-Commercially-Available-QPU.html](https://www.globenewswire.com/news-release/2023/12/06/2792153/0/en/Rigetti-Launches-the-Novera-QPU-the-Company-s-First-Commercially-Available-QPU.html), 2023.
- [25] Ernesto Palidda. Quantum feature maps and Chebyshev inequality. *IEEE Quantum Week*, 2024.
- [26] Scikit-Learn. [/https://scikit-learn.org/stable/supervised_learning.html](https://scikit-learn.org/stable/supervised_learning.html).
- [27] Scikit-Learn. [/https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html](https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html).
- [28] Scikit-Learn. [/https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.train_test_split.html](https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.train_test_split.html).
- [29] Marco Painsi, Amir Kalev, Dan Padilha, and Brendan Ruck. Estimating expectation values using approximate quantum states. *Quantum*, 5, 2021.

A Feature Embedding

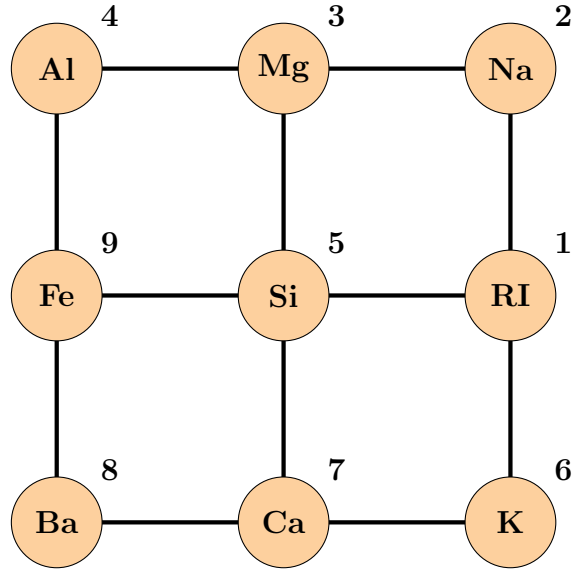


Figure 2: Embedding of a 9-qubit quantum feature map with single qubit feature encoding on the square-grid QPU graph (Ankaa/Novera). Circles indicate computational qubits and lines indicate tunable couplers. Qubits are indexed from 1 to 9 and feature names are shown inside the circles.

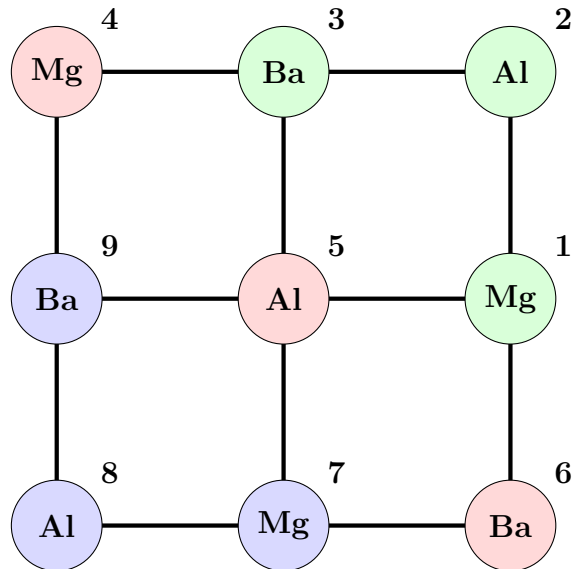


Figure 3: Embedding of a 9-qubit quantum feature map with triple qubit encoding of the most important features on the square-grid QPU graph (Ankaa/Novera). Circles indicate computational qubits and lines indicate tunable couplers. Qubits are indexed from 1 to 9 and feature names are shown inside the circles.

B Quantum Feature Map

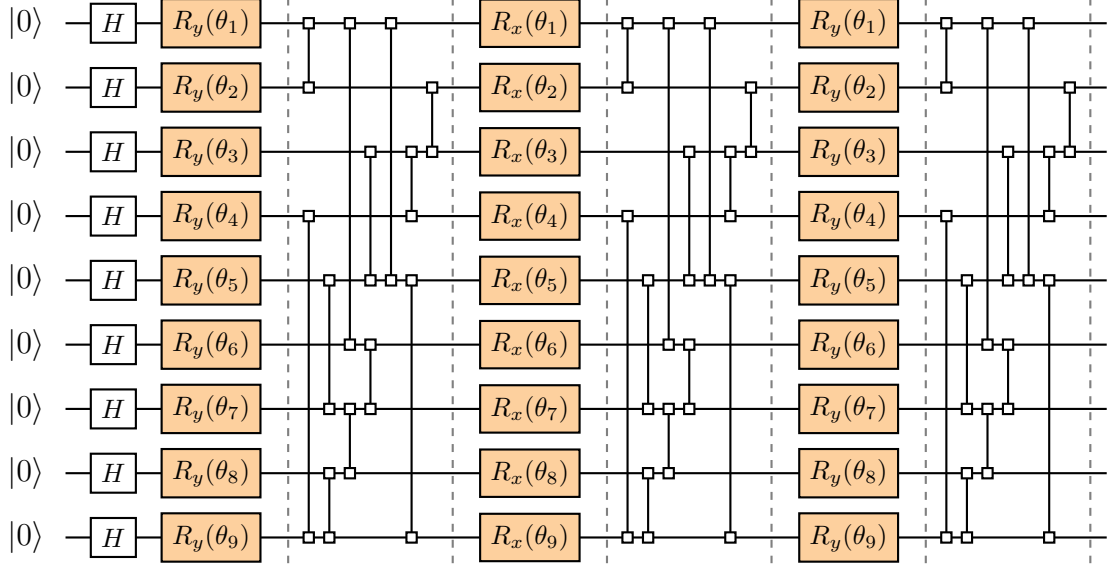


Figure 4: 9-qubit quantum circuit (quantum feature map) with single qubit feature encoding repeated three times. Coloured boxes indicate 1-qubit gates (rotations around the y and x axis) encoding the features and vertical lines indicate 2-qubit gates (i SWAP gates) creating entanglement. H gates are Hadamard gates creating equal superposition of $|0\rangle$ and $|1\rangle$ states.

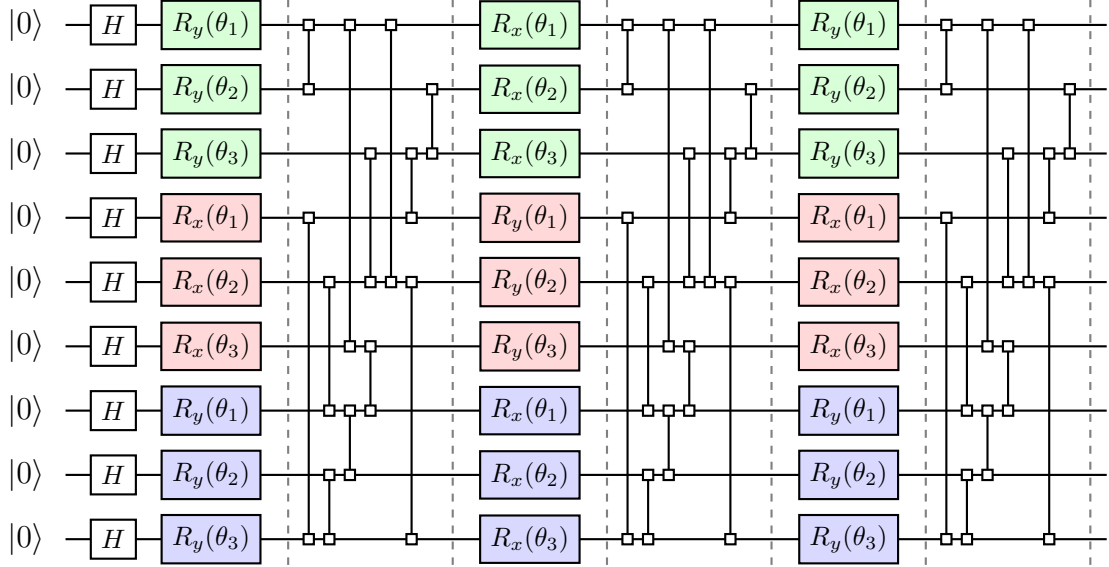


Figure 5: 9-qubit quantum circuit (quantum feature map) with triple qubit feature encoding repeated three times. Coloured boxes indicate 1-qubit gates (rotations around the y and x axis) encoding top three features and vertical lines indicate 2-qubit gates (i SWAP gates) creating entanglement. H gates are Hadamard gates creating equal superposition of $|0\rangle$ and $|1\rangle$ states.

C Estimation of Frobenius Distance on a Quantum Computer

Consider the space of linear operators $\mathcal{H}_{\mathcal{O}}$ with dimension $D_{\mathcal{O}} = D^2 = (2^n)^2$ acting in the Hilbert space $\mathcal{H}$ of a quantum system with dimension $D = 2^n$. A distance squared d^2 between two Hermitian operators p and q can be derived from a linear product $(\cdot, \cdot)$ defined in $\mathcal{H}_{\mathcal{O}}$:

$$d^2(p, q) = \|p - q\|^2 = (p - q, p - q).$$

If $\{B_i\}$ is a complete orthonormal set in $\mathcal{H}_{\mathcal{O}}$, then we can express p and q as vectors in $\mathbb{C}^{D_{\mathcal{O}}}$ with components $p_i := (B_i, p)$ and $q_i := (B_i, q)$ respectively. The Pauli operators

$$\{V_i := V_i^1 \otimes \dots \otimes V_i^n\}, \quad V_i^j = \{I, X, Y, Z\},$$

corresponding to the j -th single-qubit identity and Pauli operators, are an orthogonal set with the choice of linear product

$$(L, M) = \text{Tr}[L^\dagger M],$$

which induces the Frobenius norm. The distance squared d^2 can be expressed as the Euclidean distance in $\mathbb{C}^{D_{\mathcal{O}}}$:

$$d^2(p, q) = \sum_{i=1}^{D_{\mathcal{O}}} |p_i - q_i|^2 = \|\mathbf{p} - \mathbf{q}\|_E^2,$$

with column vectors

$$\mathbf{p} := (p_1, \dots, p_{D_{\mathcal{O}}})^\top \quad \text{and} \quad \mathbf{q} := (q_1, \dots, q_{D_{\mathcal{O}}})^\top.$$

Fixing the normalisation of the operators $\{V_i\}$, i.e., choosing $B_i = V_i/\sqrt{D}$ and p and q representing density matrices, the elements p_i and q_i become real numbers that can be expressed as

$$p_i = \frac{1}{\sqrt{D}} \text{Tr}[V_i p] = \frac{1}{\sqrt{D}} \langle V_i \rangle_p$$

and

$$q_i = \frac{1}{\sqrt{D}} \text{Tr}[V_i q] = \frac{1}{\sqrt{D}} \langle V_i \rangle_q,$$

and therefore

$$d^2(p, q) = \frac{1}{D} \sum_{i=1}^{D_{\mathcal{O}}} |\text{Tr}[V_i p] - \text{Tr}[V_i q]|^2 = \frac{1}{D} \sum_{i=1}^{D_{\mathcal{O}}} |\langle V_i \rangle_p - \langle V_i \rangle_q|^2.$$

The Frobenius distance as defined in (4) can be written as a Euclidean distance between vectors of expectation values of Paulis

$$d_F^2(p, q) = \frac{1}{4D} \sum_{i=1}^{D_{\mathcal{O}}} |\langle V_i \rangle_p - \langle V_i \rangle_q|^2 = \frac{1}{4D} \|\mathbf{P} - \mathbf{Q}\|_E^2,$$

where

$$\mathbf{P} := (\langle V_1 \rangle_p, \dots, \langle V_{D_{\mathcal{O}}} \rangle_p)^\top$$

and

$$\mathbf{Q} := (\langle V_1 \rangle_q, \dots, \langle V_{D_{\mathcal{O}}} \rangle_q)^\top.$$

The complete basis set of operators V_i consists of 4^n elements. This makes working with the complete basis set impractical or even impossible for large n . Therefore, we choose a subset of L operators V_i from the full set and approximate the sum on the expectation values of all Pauli operators with the sum on the subset of L Pauli operators:

$$\frac{1}{4D} \sum_{i=1}^L |\langle V_i \rangle_p - \langle V_i \rangle_q|^2 \leq \frac{1}{4D} \sum_{i=1}^{D_{\mathcal{O}}} |\langle V_i \rangle_p - \langle V_i \rangle_q|^2.$$

For example, L can be the set of all first order Pauli operators. For many practical applications we can ignore the higher order tensor products of Pauli operators since their expectation values tend to make negligibly small contribution.

Thus, we can define the Frobenius distance estimator as

$$\tilde{d}_F(p, q) := \frac{1}{2} \sqrt{\frac{\sum_{i=1}^L |\langle V_i \rangle_p - \langle V_i \rangle_q|^2}{D}}.$$

If we consider the subspace of $\mathcal{H}_{\mathcal{O}}$ spanned by the L Pauli operators, then

$$\tilde{d}_F(p, q) := \frac{1}{2} \sqrt{\frac{\|\mathbf{P}_L - \mathbf{Q}_L\|_E^2}{D}},$$

where

$$\mathbf{P}_L := (\langle V_1 \rangle_p, \dots, \langle V_L \rangle_p)^\top$$

and

$$\mathbf{Q}_L := (\langle V_1 \rangle_q, \dots, \langle V_L \rangle_q)^\top.$$

We can interpret this as the exact Frobenius distance with a quantum feature map that does not map to the full density matrix, but to vectors of components of the full density matrix on a Pauli basis.

Algorithm 1 provides detailed instructions for obtaining the Frobenius distance estimate on a quantum computer between the density matrices representing two datasets where we restrict ourselves to using the first order Pauli operators only.

Algorithm 1 Frobenius distance estimator for two datasets

Result: Frobenius distance.

- 1: Design n -qubit quantum circuit (quantum feature map) with layers of one-qubit gates (encoding sample) alternating with layers of two-qubit gates (creating entanglement). By default, the range of rotation angles should be set at $[0, \pi/m]$, where m is the number of data encoding layers.
- 2: Prepare two datasets (A and B) – calculate features, standardise features, etc.
- 3: For dataset A :
 - For each sample $i = 1, \dots, [A]$, run the quantum circuit N times measuring Pauli Z_k on all quantum registers $k = 1, \dots, n$ (+1 if we measure $|0\rangle$ and -1 if we measure $|1\rangle$). Calculate expectation value $\langle Z_k \rangle_i^A$ as the average value of measured Z_k across all quantum circuit runs.
 - For each sample $i = 1, \dots, [A]$, place H gate on **all** quantum registers before measurement and run the quantum circuit N times measuring Pauli X_k on all quantum registers $k = 1, \dots, n$ (+1 if we measure $|0\rangle$ and -1 if we measure $|1\rangle$). Calculate expectation value $\langle X_k \rangle_i^A$ as the average value of measured X_k across all quantum circuit runs.
 - For each sample $i = 1, \dots, [A]$, place $HS^\dagger$ gates on **all** quantum registers before measurement and run the quantum circuit N times measuring Pauli Y_k on all quantum registers $k = 1, \dots, n$ (+1 if we measure $|0\rangle$ and -1 if we measure $|1\rangle$). Calculate expectation value $\langle Y_k \rangle_i^A$ as the average value of measured Y_k across all quantum circuit runs.
- 4: For dataset B :
 - For each sample $j = 1, \dots, [B]$, run the quantum circuit N times measuring Pauli Z_k on all quantum registers $k = 1, \dots, n$ (+1 if we measure $|0\rangle$ and -1 if we measure $|1\rangle$). Calculate expectation value $\langle Z_k \rangle_j^B$ as the average value of measured Z_k across all quantum circuit runs.
 - For each sample $j = 1, \dots, [B]$, place H gate on **all** quantum registers before measurement and run the quantum circuit N times measuring Pauli X_k on all quantum registers $k = 1, \dots, n$ (+1 if we measure $|0\rangle$ and -1 if we measure $|1\rangle$). Calculate expectation value $\langle X_k \rangle_j^B$ as the average value of measured X_k across all quantum circuit runs.
 - For each sample $j = 1, \dots, [B]$, place $HS^\dagger$ gates on **all** quantum registers before measurement and run the quantum circuit N times measuring Pauli Y_k on all quantum registers $k = 1, \dots, n$ (+1 if we measure $|0\rangle$ and -1 if we measure $|1\rangle$). Calculate expectation value $\langle Y_k \rangle_j^B$ as the average value of measured Y_k across all quantum circuit runs.
- 5: We now have two arrays: $[A] \times 3n$ array of expectation values of first order Paulis for dataset A and $[B] \times 3n$ array of expectation values of first order Paulis for dataset B . For every column of these arrays we calculate the average expectation values across all samples in the datasets:

$$\overline{\langle X_1 \rangle^A} = \frac{1}{[A]} \sum_{i=1}^{[A]} \langle X_1 \rangle_i^A, \dots; \quad \overline{\langle X_1 \rangle^B} = \frac{1}{[B]} \sum_{j=1}^{[B]} \langle X_1 \rangle_j^B, \dots$$

6: We now have two $3n$ -dimensional column vectors for datasets A and B :

$$\left(\overline{\langle X_1 \rangle^A}, \overline{\langle X_2 \rangle^A}, \dots\right)^\top; \quad \left(\overline{\langle X_1 \rangle^B}, \overline{\langle X_2 \rangle^B}, \dots\right)^\top.$$

The next step is to calculate the column vector of differences:

$$\mathbf{c} = \left(\overline{\langle X_1 \rangle^A} - \overline{\langle X_1 \rangle^B}, \overline{\langle X_2 \rangle^A} - \overline{\langle X_2 \rangle^B}, \dots\right)^\top.$$

7: Calculate the estimate of the Frobenius distance:

$$\tilde{d}_F(A, B) = \frac{1}{2} \sqrt{\frac{\mathbf{c}^\top \mathbf{c}}{2^n}}.$$

Remark: Note that, in practice, separate executions under different changes of bases can be avoided by utilising the approximate state method [29].